

Mathematics → 2

Statistics

Unit - I, II

Probability

Unit - I

→ Measure of central tendency

Mean

Median

Mode

→ Mean

The arithmetic mean of $x_1, x_2, \dots, x_n$ is defined as $\bar{x} = \frac{1}{n} (x_1 + x_2 + \dots + x_n) = \frac{1}{n} \sum_{i=1}^n x_i$

If the data has frequencies

Variable (x) $x_1 \quad x_2 \quad \dots \quad x_n$

Frequency (f) $f_1 \quad f_2 \quad \dots \quad f_n$

then

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n f_i x_i$$

In case variable is divided into class intervals then $x_i \rightarrow$ mid value of each class interval

→ Theorem

If two variables x & y are related by equation

$y = \frac{x-c}{d}$ then and $\bar{y} = \frac{\bar{x}-c}{d}$ where c & d are any numbers.

$$\text{If } y = \frac{x-c}{d} \Rightarrow \bar{y} = \frac{\bar{x}-c}{d}$$



Q Find mean

Weights (in g)	110-119	120-129	130-139	140-149	150-159
Frequency	5	7	12	20	16
	160-169	170-179	180-189		
	10	7	3		

C I	f	x_i	$d_i = x_i - 144.5$	$f d_i$
109.5 - 119.5	5	114.5	-30	-150
119.5 - 129.5	7	124.5	-20	-140
129.5 - 139.5	12	134.5	-10	-120
139.5 - 149.5	20	144.5	0	0
149.5 - 159.5	16	154.5	10	160
159.5 - 169.5	10	164.5	20	200
169.5 - 179.5	7	174.5	30	210
179.5 - 189.5	3	180.5	40	120
	<u>$\Sigma f_i = 80$</u>			<u>280</u>

$$\bar{x} = a + \frac{\Sigma f d_i}{\Sigma f_i}$$

$$\bar{x} = 144.5 + \frac{280}{80}$$

$$\bar{x} = 144.5 + 3.5$$

$$\bar{x} = 148 \text{ g}$$

→ values badi h to chota krlo

$$\bar{x} = a + \frac{\Sigma f d_i}{\Sigma f_i}$$

$$\bar{x} = a + h \left(\frac{\Sigma f_i u_i}{\Sigma f_i} \right)$$

$$d_i = x_i - a$$

$$u_i = \frac{x_i - a}{h}$$

$a \rightarrow$ expected central value of x

$h \rightarrow$ class size

Q Avg marks obtd by class of 70 students was found to be 65. Later its found marks of 1 student were wrongly recorded as 85 in place of 58. Find correct mean.

$$65 = \frac{\sum x_i}{70}$$

$$\sum x_i = 65 \times 70 = 4550$$

$$\sum x_i' = \sum x_i - 85 + 58 = 4550 - 27 = 4523$$

$$\Rightarrow \bar{x} = \frac{4523}{70} = \frac{646.142}{10} = 64.61$$

Q

CI	f
3-5	32
5-7	f_1
7-9	57
9-11	f_2
11-13	25
	<u>200</u>

mean = 7.74
 f_1 & $f_2 = ?$

CI	x_i	f_i	$f_i x_i$
3-5	4	32	128
5-7	6	f_1	$6f_1$
7-9	8	57	456
9-11	10	f_2	$10f_2$
11-13	12	25	300

$$200 = 114 + f_1 + f_2$$

$$f_1 + f_2 = 86$$

$$884 + 6f_1 + 10f_2$$

$$884 + 6(86 - f_2) + 10f_2$$

$$884 + 14f_2 + 516$$

$$4f_2 + 1400$$

$$\Rightarrow \frac{4f_2 + 1400}{200} = 7.74$$

$$4f_2 + 1400 = 1548$$

$$4f_2 = 148$$

$$f_2 = 37$$

$$f_1 = 86 - 37 = 49$$



→ Median

It's the middle most value when observations when arranged in $\uparrow$ or $\downarrow$ order of magnitude after arranging n values in $\uparrow$ or $\downarrow$ order

- If n is odd $\rightarrow \left(\frac{n+1}{2}\right)^{\text{th}}$ observation is median
- If n is even $\rightarrow \frac{1}{2} \left[\left(\frac{n}{2}\right)^{\text{th}} + \left(\frac{n+1}{2}\right)^{\text{th}} \right]$ observation is median
- For data with frequency
Find the commulative freq N
Median = obs having $f \geq \frac{N+1}{2}$
- For grouped frequency distribution
Median = $l + h \left(\frac{N/2 - cf}{f} \right)$

Median class $\rightarrow$ having freq $\geq cf$

$l \rightarrow$ lower limit of median class

$cf \rightarrow$ commulative freq of class preceding the median class

f & $h \rightarrow$ frequency & class size of median class

Q. Height	f_i	cf	$N/2 = 29.5$
59.5 - 63.5	8	8	
63.5 - 67.5	3	11	
67.5 - 71.5	18	29	$\rightarrow$ median class
71.5 - 75.5	6	35	$md = 71.5 + 4 \left[\frac{29.5 - 29}{6} \right]$
75.5 - 79.5	16	51	
79.5 - 83.5	8	59	$= 71.83$
		$N = 59$	



→ Mode

The observation having maximum frequency is called mode.

• For grouped frequencies

$$\text{Mode} = l + \left[\left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h \right]$$

Modal class → class with highest freq.

f_1 → freq of modal class

f_0 → freq of class before modal class

f_2 → freq of class after modal class

Height	No of students
10-15	6
15-20	9 → f_0
20-25	11 → modal class
25-30	7 → f_2
30-35	7

$$\text{Mode} = 20 + 5 \left[\frac{11-9}{22-16} \right]$$

$$= 20 + 5 \left[\frac{2}{6} \right]$$

$$= 20 + 1.66$$

$$= 21.66$$

Q	CI	f	cf
	0-10	3	3
	10-20	a	3+a
	20-30	20	23+a
	30-40	12	35+a
	40-50	b	35+a+b

Median lies in median class

$\Rightarrow$ Median class is 20-30

$$27 = 20 + 10 \left[\frac{\frac{35+a+b}{2} - 3-a}{20} \right]$$

$$7 = \frac{35+a+b - 6 - 2a}{2 \times 2}$$

$$28 = 35 - 6 + b - a$$

$$-1 = b - a$$

$$a - b = 1 \quad \text{--- (1)}$$

Modal class $\rightarrow$ 20-30

$$26 = 20 + 10 \left[\frac{20-a -}{40-a-12} \right]$$

$$\frac{3}{5} = \left[\frac{20-a}{28-a} \right]$$

$$84 - 3a = 100 - 5a$$

$$2a = 16$$

$$a = 8$$

$$\Rightarrow b = a - 1$$

$$b = 7$$



→ Variance & Standard deviation

Mean of squares of differences of observations from their mean is variance.

The positive sq
root of variance is standard deviation (σ)
if $x_1, x_2, \dots, x_n$ is data then

$$\text{Var}(x) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\sigma = +\sqrt{\text{Var}(x)}$$

- For ungrouped frequency data

$$x_1, x_2, \dots, x_n$$

$$t_1 \quad t_2 \quad \dots \quad t_n$$

$$\text{Var}(x) = \frac{1}{n} \sum_{i=1}^n f_i (x_i - \bar{x})^2 \quad \& \quad \sigma = +\sqrt{\text{Var}(x)}$$

- If 2 variables x & y are related by
eq $y = x - c$ then $\sigma_y = \sigma_x$
sol sd

→ Theorem

If $x_1, x_2, \dots, x_n$ be n data then

$$\text{var}(x) = \frac{1}{n} \sum_{i=1}^n x_i^2 - \left(\frac{1}{n} \sum_{i=1}^n x_i \right)^2$$

or for freq :-

for freq.:-

$$\text{Var}(x) = \frac{1}{N} \sum_{i=1}^n f_i x_i^2 - \left(\frac{1}{N} \sum_{i=1}^n f_i x_i \right)^2$$

can be proved using probability

$$\begin{aligned}\text{var}(x) &= E[(x - E(x))^2] \\ &= E[x^2 + E^2(x) + 2xE(x)]\end{aligned}$$

↓
expectation

$$\therefore E[E(x)] = x$$

Page

Date



$$= E(x^2) - E(2xE(x)) + E(E^2(x))$$

$$= E(x^2) - 2E(x)E(x) + E^2(x)$$

$$= E(x^2) - 2E^2(x) + E^2(x)$$

$$= E(x^2) - E^2(x)$$

$$= \frac{1}{n} \sum x_i^2 - \left(\frac{1}{n} \sum x_i \right)^2$$

Q Variation of 20, 85, 120, 60, 40

$$\bar{x} = \frac{20 + 85 + 120 + 60 + 40}{5} = 65$$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
20	-45	2025
85	+20	400
120	55	3025
60	-5	25
40	-25	625
		<u>6100</u>

$$\frac{\sum (x_i - \bar{x})^2}{N} = \frac{6100}{5} = 1220$$

$$\text{Var}(x) = 1220$$

$$\sigma = \sqrt{1220} = 34.93$$

Q

x_i	f_i	$d = x_i - 64.5$	$f_i d_i$	$f_i d_i^2$
94.5	2	30	60	1800
84.5	12	20	240	4800
74.5	22	10	220	2200
64.5	20	0	0	0
54.5	14	-10	-140	+1400
44.5	4	-20	-80	+1600
34.5	1	-30	-30	900
	<u>75</u>		<u>270</u>	<u>12700</u>



$$\text{Var}(d) = \frac{12700}{75} - \left(\frac{270}{75}\right)^2$$

$$= 169.33 - (3.6)^2$$

$$= 156.37$$

$$\text{Now, } \sigma_d = \sqrt{156.37} = 12.5047$$

$$\therefore \sigma_x = 1 \times \sigma_d = 12.5047$$

Q	Weights	f	x_i	$x_i - 53$	$fidi$	$fidi^2$
	35.5 - 40.5	14	38	-15	-210	3150
	40.5 - 45.5	26	43	-10	-260	2600
	45.5 - 50.5	40	48	-5	-200	1000
	50.5 - 55.5	33	53	0	0	0
	55.5 - 60.5	50	58	5	250	1250
	60.5 - 65.5	37	63	10	370	3700
	65.5 - 70.5	25	68	15	375	5625
		<u>225</u>			<u>325</u>	<u>17325</u>

$$\text{Var}(d) = \frac{17325}{225} - \left(\frac{325}{225}\right)^2$$

$$= 77 - (1.45)^2$$

$$= 77 - 2.1025$$

$$= 74.8975$$

$$\sigma_d = 1 \times \sigma_x = \sqrt{74.8975}$$

$$\sigma_x = 8.655$$

1. coefficient of variation (C.V.)
- Measure of central tendency represents set of all observation/data of variable while σ shows measure of consistency of data. To compare the consistency of data, we find CV. The variable having greater CV is lesser consistent.
- $$CV = \frac{\text{Standard deviation}}{\text{Mean}} \times 100$$

$$CV = \frac{\sigma}{\bar{x}} \times 100$$

CV $\uparrow\uparrow$ consistency of variable $\downarrow\downarrow$

Q. which batter is better run scorer?
Scores of batsman A & B in 10 innings are

A	B	$x_A - \bar{x}$	$(x_A - \bar{x})^2$	$x_B - \bar{x}$	$(x_B - \bar{x})^2$
19	32	-31	961	-14	196
31	28	-19	361	-18	324
48	47	-2	4	1	1
53	63	3	9	17	289
67	71	17	289	25	625
90	39	40	1600	-7	49
10	10	-40	1600	-36	1296
62	60	12	144	14	196
40	96	-10	100	50	2500
80	14	30	900	-32	1024
<u>500</u>	<u>460</u>		<u>5971</u>		<u>6500</u>

$$\bar{x}_A = \frac{500}{10} = 50 \quad \text{and} \quad \bar{x}_B = \frac{460}{10} = 46$$

$$\text{Var}(A) = \frac{5971}{10} = 597.1 \quad \text{Var}(B) = 650$$

$$\sigma_A = 24.43 \quad \& \quad \sigma_B = 25.49$$

NOW, $CV_A = \frac{24.43 \times 100}{50}$ & $CV_B = \frac{25.49 \times 100}{46}$

$CV_A = \frac{0.488 \times 100}{48.8}$ $CV_B = \frac{0.554 \times 100}{55.4}$

$\Rightarrow CV_B > CV_A$

consistency of B $\downarrow \downarrow$

$\Rightarrow$ A is better run scorer.

$\rightarrow$ Moments

The k^{th} moment of values $x_1, x_2, \dots, x_n$ about a number 'a' is defined as

$= \frac{1}{n} [(x_1 - a)^k + (x_2 - a)^k + \dots + (x_n - a)^k]$

$= \frac{1}{n} \sum_{i=1}^n (x_i - a)^k$

• For frequency distributions

$\frac{1}{n} \sum_{i=1}^n f_i (x_i - a)^k$

$\rightarrow$ Central moments and raw moments

Any moment about mean is called central moment and is denoted by μ^k

Any moment about zero is called raw moment and denoted by μ'^k

→ Raw moment ($a=0$)

$$\mu_0' = 1$$

$$\mu_1' = \frac{1}{n} \sum (x_i - 0)^1 = \frac{\sum x_i}{n} = \text{mean} = \mu_1'$$

$$\mu_2' = \frac{1}{n} \sum (x_i)^2 = \mu_2'$$

→ Central moment ($a = \bar{x}$)

$$\mu_1 = \frac{1}{n} \sum (x_i - \bar{x})$$

$$\mu_2 = \frac{1}{n} \sum (x_i - \bar{x})^2 = \text{variance} = \mu_2$$

→ Prove $\mu_1 = 0$

$$\mu_1 = \frac{1}{n} \sum x_i - \bar{x}$$

$$= \frac{1}{n} \sum x_i - \frac{1}{n} \sum \bar{x}$$

$$= \text{Mean} - \frac{1}{n} \sum \bar{x}$$

$$= \bar{x} - \text{Mean}$$

$$= \bar{x} - \bar{x}$$

$$= 0$$

★ → Relation b/t central moment & any moment

$$\mu_0 = 1$$

$\mu_r' \rightarrow$ about any no.

$$\mu_1 = 0$$

$$\mu_2 = \mu_2' - (\mu_1')^2$$

$$\mu_3 = \mu_3' - 3\mu_2' \mu_1' + 2(\mu_1')^3$$

$$\mu_4 = \mu_4' - 4\mu_3' \mu_1' + 6\mu_2' (\mu_1')^2 - 3(\mu_1')^4$$

} learn

$$\text{if } y = \frac{x - c}{d} \text{ then } \mu_r(y) = \frac{\mu_r(x)}{(d)^r}$$

Q First four moments about 5 are 2, 10, 40, 218
Find first 4 central moments & moments about 0

$$\mu_1 = 0$$

$$\mu_2 = 10 - (2)^2 = 10 - 4 = 6$$

$$\mu_3 = 40 - 3(10)(2) + 2(2)^3 = 40 - 60 + 16 = -4$$

$$\mu_4 = 218 - 4(40)(2) + 6(10)(2)^2 - 3(2)^4$$

$$\mu_4 = 218 - 320 + 240 - 48$$

$$\mu_4 = 218 - 80 - 48 = 218 - 128 = 90$$

Now

$$\Rightarrow \text{First moment about 5} = 2$$

$$\Rightarrow \sum_{i=1}^n (x_i - 5) = 2$$

$$\Rightarrow \frac{\sum x_i}{n} = 2 + \frac{5n}{n} = 7$$

$$\Rightarrow \text{Mean} = 7$$

$$\Rightarrow \mu_1' = 7$$

$$\Rightarrow \mu_2 = \mu_2' - (\mu_1')^2$$

$$\Rightarrow 6 = \mu_2' - 49$$

$$\Rightarrow \mu_2' = 55$$

$$\Rightarrow -4 = \mu_3' - 3(55)(7) + 2(343)$$

$$\Rightarrow -4 = \mu_3' - 465$$

$$\Rightarrow \mu_3' = 465$$

$$\Rightarrow 90 = \mu_4' - 4(465)(7) + 6(55)(49) - 3(49)(49)$$

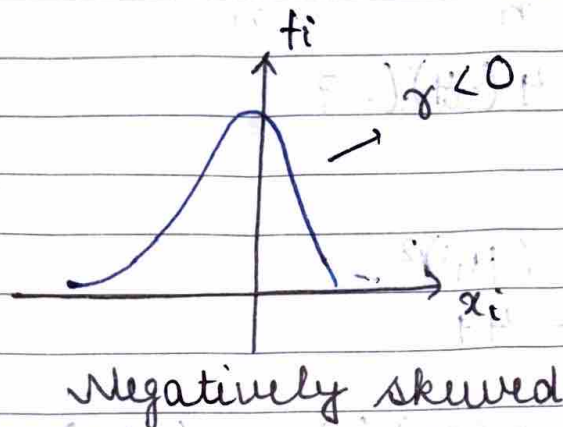
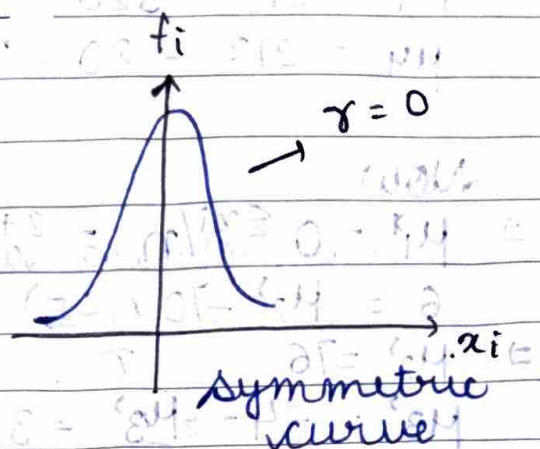
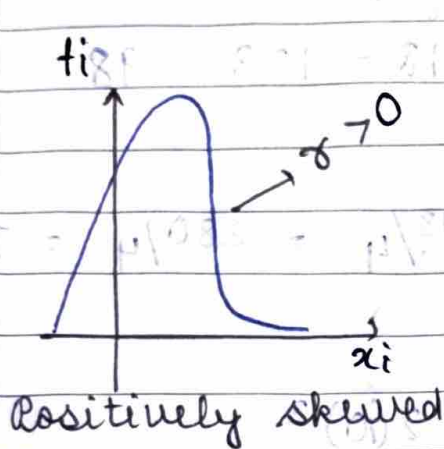
$$\Rightarrow 90 = \mu_4' - 13,020 + 16,170 - 7203$$

$$\Rightarrow \mu_4' = 4143$$

→ Skewness → Lack of symmetry
skewness of a frequency distribution are

Tells about asymmetry
coeff. of skewness (β_1) = $\frac{\mu_3^2}{\mu_2^3}$

$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}}$



→ Kurtosis

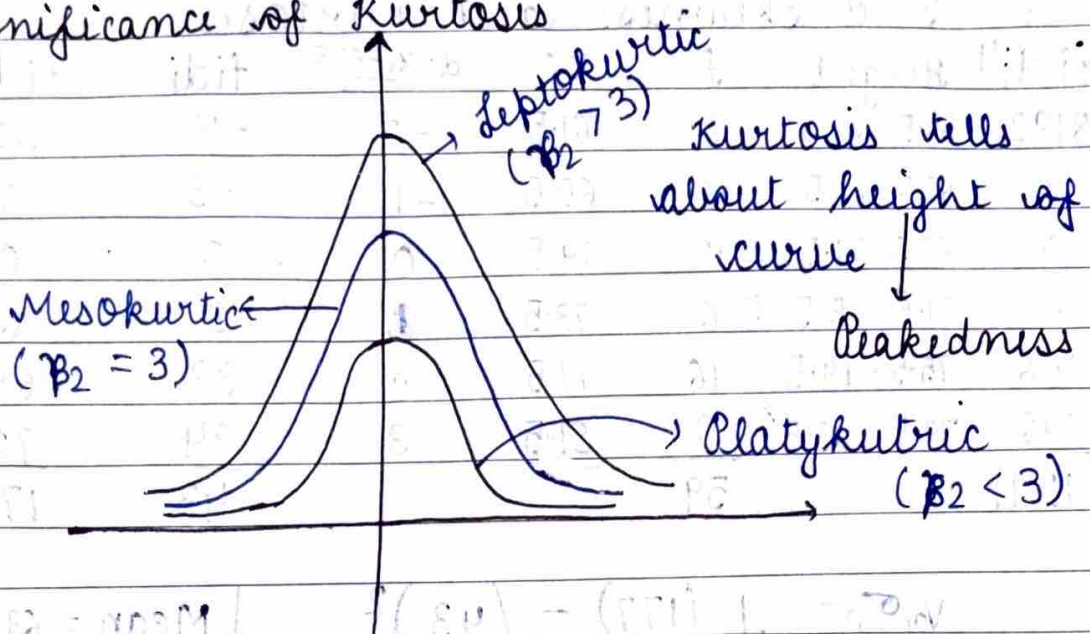
coeff of kurtosis (β_2) = $\frac{\mu_4}{\mu_2^2}$

$\beta_2 = \frac{\mu_4}{\mu_2^2}$

$\gamma_2 = \beta_2 - 3$



→ Significance of Kurtosis



- Kurtosis of normal variate is zero
- 1. Peakedness of a normal distribution is 3. So if peakedness is less than 3 then kurtosis will be negative and is called (platykurtic)
- 2. If peakedness is greater than 3 then kurtosis will be positive & be called leptokurtic
- 3. If peakedness is 3, kurtosis is 0. This type of distribution is mesokurtic

→ Other formulas to find skewness

$$\text{Skewness} = \frac{\text{Mean} - \text{Mode}}{\sigma}$$

$$\text{Skewness} = \frac{3(\text{Mean} - \text{Median})}{\sigma}$$



Q Find skewness and kurtosis

Huight	f	x_i	$d_i = (x_i - 69.5)4$	fid_i	fid_i^2	fid_i^3	fid_i^4
59.5-63.5	8	61.5	-2	-16	32	-64	128
63.5-67.5	3	65.5	1	-3	3	-3	3
67.5-71.5	18	69.5	0	0	0	0	0
71.5-75.5	6	73.5	1	6	6	6	6
75.5-79.5	16	77.5	2	32	32	128	256
79.5-83.5	8	81.5	3	24	24	216	648
	59			43	177	283	1041

Moments about 69.5:-

$$\mu_1' = \frac{(43)}{59} = 2.915$$

$$\mu_2' = \frac{4 \times 177}{59} = 3 \times 16 = 48$$

$$\mu_3' = \frac{283 \times 4^3}{59} = 4.79 \times 64 = 306.56$$

$$\mu_4' = \frac{4 \times 1041}{59} = 17.64 \times 4^4 = 4516.88$$

Central moments

$$\mu_1 = 0$$

$$\mu_2 = \mu_2' - \mu_1'^2 = 48 - (2.915)^2 = 39.53$$

$$\mu_3 = 306.56 - 3(2.915)(48) + 2(2.915)^3$$

$$\mu_3 = -63.19$$

$$\mu_4 = 4516.88 - 4(306.56)(2.915) + 6(48)(2.915)^2 - 3(2.915)^4$$

$$\mu_4 = 3172.208$$

$$\beta_2 = 2.03$$

$$\gamma_2 = -0.97$$

$$\beta_2 < 3$$

Platykurtic

$$\beta_1 = 0.064$$

$$\gamma_1 = -0.252 = \frac{\mu_3}{\mu_2^{3/2}}$$

↓

negatively skewed



→ Bivariate data

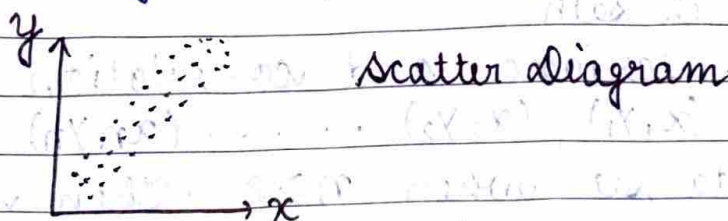
Let x and y be 2 random variables defined on an event space S of an experiment. Then the pair (x, y) is called bivariate on same event space. These pair of values are called as bi-variate data.

- Both discrete variables
- one continuous and 1 discrete variable
- Both continuous variables

→ Types of co-relation

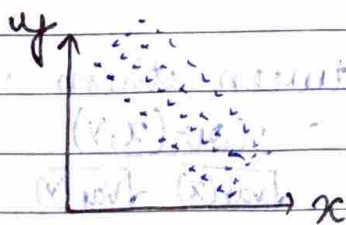
- Positively co-related (directly)

If y has tendency to increase as x increases then x and y are positively co-related.



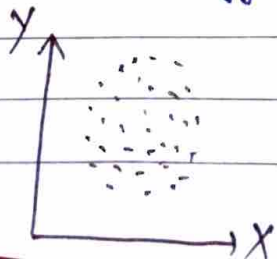
- Negatively co-related (inversely)

If y has tendency to decrease as x increase we say x & y are negatively co-related.



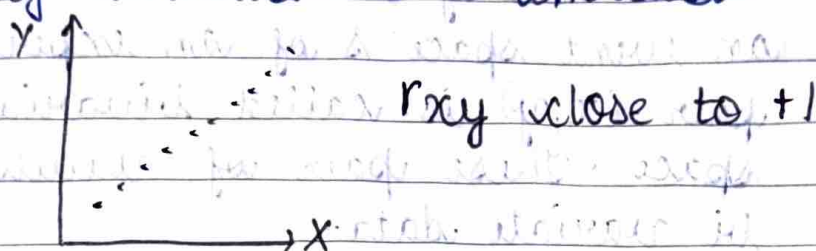
- Uncorrelated or zero co-related

If value of y aren't affected by changes of value in x

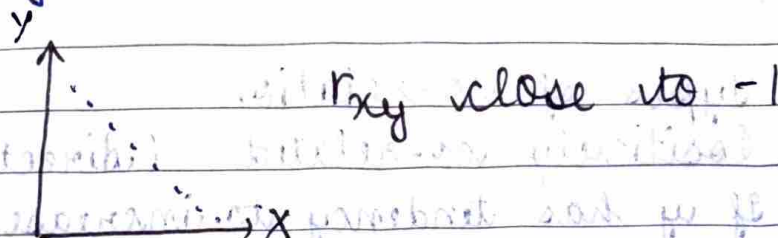




- strictly positively correlated
if y strictly increase as x increases



- strictly negatively co-related
if y strictly decrease as x increases



2 variables ka variance
↑ ek sath

- co-variance and co-relation

Let $(x_1, y_1), (x_2, y_2) \dots (x_n, y_n)$ be a bivariate data set when $n \geq 2$. Then covariance between variables x & y is defined as

$$\text{cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{n} \sum_{i=1}^n x_i y_i - \bar{x} \bar{y}$$

To solve ques.

The co-relation between them is defined as

$$r_{xy} = \rho_{xy} = \text{corr}(x, y) = \frac{\text{cov}(x, y)}{\sqrt{\text{Var}(x)} \sqrt{\text{Var}(y)}} = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$$

correlation coefficient



→ Properties of covariance

Let $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ be a bivariate data set. Let $a, b, c, d \in \mathbb{R}$ be any numbers where $a, c \neq 0$. Then:-

1. $\text{cov}(ax+b, cy+d) = ac \text{cov}(x, y)$
2. $\text{cov}(x, y) = \text{cov}(y, x)$
3. $\text{cov}(x, x) = \text{var}(x)$
4. $\text{cov}(ax+by, cx+dy) = ac \text{var}(x) + (ad+bc) \text{cov}(x, y) + bd \text{var}(y)$
5. $\text{var}(x+y) = \text{var}(x) + 2 \text{cov}(x, y) + \text{var}(y)$
6. $\text{cov}(x+y, x-y) = \text{var}(x) - \text{var}(y)$

→ Properties of correlation coefficient

1. $r_{xy} = r_{yx}$
2. $\text{cov}(x, y)$ and σ_x, σ_y have same units
3. Greater value of r_{xy} shows strength of association between x & y is greater
4. Let (x, y) and (u, v) represent two sets of bivariate data such that $u = ax+b$ and $v = cy+d$.
then $r_{uv} = \frac{ac}{|a||c|} r_{xy}$
- If a and c have same sign, then $r_{uv} = r_{xy}$
- If a and c have opposite sign, then $r_{uv} = -r_{xy}$
5. $\text{var}(x+y) = \text{var}(x) + 2\sigma_x \sigma_y r_{xy} + \text{var}(y)$
6. $\text{var}(x-y) = \text{var}(x) - 2\sigma_x \sigma_y r_{xy} + \text{var}(y)$
7. $-1 \leq r_{xy} \leq 1$
8. If $r_{xy} = \pm 1$ (the max and min value of coeff) then y is the linear function of x & vice versa

→ If $u = ax+b$ and $v = cy+d$ P.T. $r_{uv} = \frac{ac}{|a||c|} r_{xy}$
 since $u = ax+b$ & $v = cy+d$
 $\Rightarrow \bar{u} = a\bar{x} + b$ & $\bar{v} = c\bar{y} + d$
 $\bar{x} = \frac{\bar{u} - b}{a}$ & $\bar{y} = \frac{\bar{v} - d}{c}$



$$\text{var}(u) = \frac{1}{n} \sum (u_i - \bar{u})^2 = \frac{1}{n} \sum [ax_i + b - (a\bar{x} + b)]^2$$

$$= \frac{1}{n} \sum [a(x_i - \bar{x})]^2 = a^2 \text{var}(x) \quad \left. \begin{array}{l} \sigma_u = |a| \sigma_x \\ \sigma_w = |c| \sigma_y \end{array} \right\}$$

$$\text{|| by } \text{var}(v) = c^2 \text{var}(y)$$

$$\text{cov}(u, v) = \frac{1}{n} \sum u_i v_i - \bar{u} \bar{v}$$

$$= \frac{1}{n} \sum [(ax_i + b)(cy_i + d) - (a\bar{x} + b)(c\bar{y} + d)]$$

$$= \frac{1}{n} \sum [acx_i y_i + adx_i + bcy_i + bd - ac\bar{x}\bar{y} - ad\bar{x} - bc\bar{y} - bd]$$

$$= \frac{1}{n} \sum [ac(x_i y_i - \bar{x}\bar{y}) + ad(x_i - \bar{x}) + bc(y_i - \bar{y})]$$

$$= ac \text{cov}(x, y) + \underbrace{ad \frac{1}{n} \sum x_i - \bar{x}}_{=0} + \underbrace{bc \frac{1}{n} \sum y_i - \bar{y}}_{=0}$$

$$\frac{1}{n} \sum x_i - \bar{x} = \frac{\sum x_i}{n} - \frac{n\bar{x}}{n} = \bar{x} - \bar{x} = 0$$

$$\Rightarrow \text{cov}(u, v) = ac \text{cov}(x, y)$$

$$\text{Hence } r_{uv} = \frac{\text{cov}(u, v)}{\sigma_u \sigma_v}$$

$$= \frac{ac \text{cov}(x, y)}{|a| |c| \sigma_x \sigma_y}$$

$$r_{uv} = \frac{ac}{|a| |c|} r_{xy}$$

$$\rightarrow \text{PT } \text{var}(x+y) = \text{var}(x) + 2\sigma_x\sigma_y r_{xy} + \text{var}(y)$$

$$\text{var}(x-y) = \text{var}(x) - 2\sigma_x\sigma_y r_{xy} + \text{var}(y)$$

$$\text{var}(x+y) = \text{var}(x) + 2\text{cov}(x, y) + \text{var}(y)$$

$$\text{var}(x+y) = \frac{1}{n} \sum [(x_i + y_i) - (\bar{x} + \bar{y})]^2$$

$$= \frac{1}{n} \sum [(x_i - \bar{x}) + (y_i - \bar{y})]^2$$

$$= \frac{1}{n} \sum (x_i - \bar{x})^2 + \frac{1}{n} \sum (y_i - \bar{y})^2 + \frac{2}{n} \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$= \text{var}(x) + \text{var}(y) + 2\sigma_x\sigma_y r_{xy}$$

$= \text{cov}(x, y)$

$$\Rightarrow \text{var}(x) + \text{var}(y) + 2\text{cov}(x, y)$$

Hence Proved

//ly

$$\text{var}(x-y) = \text{var}(x) - 2\sigma_x\sigma_y r_{xy} + \text{var}(y)$$

$$\rightarrow \text{PT } -1 \leq r_{xy} \leq 1$$

Let (u, v) represent a bivariate data such that

$$u = \frac{x - \bar{x}}{\sigma_x} \quad v = \frac{y - \bar{y}}{\sigma_y}$$

$$\Rightarrow \bar{u} = \frac{\bar{x} - \bar{x}}{\sigma_x}$$

$$\& \bar{v} = \frac{\bar{y} - \bar{y}}{\sigma_y}$$

$$\bar{u} = 0 \quad \& \quad \bar{v} = 0$$

$$\Rightarrow u_i = \frac{x_i - \bar{x}}{\sigma_x} \quad \text{and} \quad v_i = \frac{y_i - \bar{y}}{\sigma_y}$$

$$\begin{aligned}
 \Rightarrow \text{cov}(u, v) &= \frac{1}{n} \sum u_i v_i - \bar{u} \bar{v} \\
 &= \frac{1}{n} \sum \left(\frac{x_i - \bar{x}}{\sigma_x} \right) \left(\frac{y_i - \bar{y}}{\sigma_y} \right) - 0 \\
 &= \frac{\left[\frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y}) \right]}{\sigma_x \sigma_y} = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y} = r_{xy}
 \end{aligned}$$

$$\Rightarrow \text{cov}(u, v) = r_{xy}$$

$$\text{Now, } \frac{1}{n} \sum u_i^2 = \frac{1}{n} \sum \frac{(x_i - \bar{x})^2}{\text{var}(x)} = \frac{\text{var}(x)}{\text{var}(x)}$$

$$\Rightarrow \frac{1}{n} \sum u_i^2 = 1$$

$$\text{Also, } \frac{1}{n} \sum v_i^2 = 1$$

Since $(u_i + v_i)^2$ is non negative
 $(u_i + v_i)^2 \geq 0$

$$\Rightarrow \sum u_i^2 + \sum v_i^2 + 2 \sum u_i v_i \geq 0$$

$$\Rightarrow \frac{\sum u_i^2}{n} + \frac{\sum v_i^2}{n} + \frac{2}{n} \sum u_i v_i \geq 0$$

$$\Rightarrow 2 + \frac{2}{n} \sum u_i v_i \geq 0$$

$$\Rightarrow 1 + \frac{1}{n} \sum u_i v_i \geq 0$$

$$\frac{1}{n} \sum u_i v_i \geq -1$$



$$\Rightarrow \text{cov}(u, v) = \frac{1}{n} \sum (u_i - \bar{u})(v_i - \bar{v}) = \frac{1}{n} \sum u_i v_i$$

$$\Rightarrow \text{cov}(u, v) \geq -1$$

$$\text{or } r_{xy} \geq -1$$

$$\text{Hly } (u_i - v_i)^2 \geq 0$$

$$\text{Similarly } r_{xy} \leq 1$$

$$\Rightarrow -1 \leq r_{xy} \leq 1$$

Q Find r_{xy}

x	y	$u = x - 67$	$v = y - 65$	uv	u^2	v^2
65	67	-2	2	-4	4	4
66	68	-1	3	-3	1	9
67	(65)	0	0	0	0	0
(67)	68	0	3	0	0	9
68	72	1	7	7	1	49
69	72	2	7	14	4	49
70	69	3	4	12	9	16
72	71	5	6	30	25	36
		8	32	32	44	172

$$\bar{u} = 8/8 = 1$$

$$\bar{v} = 32/8 = 4$$

$$\text{cov}(u, v) = \frac{1}{n} \sum u_i v_i = \frac{1}{n} (32)$$

$$= \frac{1}{8} (32) = \frac{4}{1}$$

$$= \frac{28}{8} = \frac{7}{2} = 3.5$$

$$\text{cov}(x, y) = 1 \times 3.5 = 3.5$$



$$\text{var}(u)$$

$$= \frac{1}{n} \sum x_i^2 - \left(\frac{\sum x_i}{n} \right)^2$$

$$= \frac{1}{8} \times 44 - 1$$

$$= 5.5 - 1 = 4.5$$

$$\sigma_u = \sigma_x = \sqrt{4.5} = 2.12$$

$$\text{var}(v)$$

$$= \frac{1}{8} \times 172 - 16 = 21.5 - 16$$

$$= 5.5$$

$$\sigma_v = \sigma_y = \sqrt{5.5} = 2.34$$

$$\Rightarrow r_{xy} = \frac{3.5}{2.12 \times 2.34}$$

$$= 0.7055$$

$$Q4 \quad u + 3x = 5 \quad 2y - v = 7 \quad r_{xy} = 0.12$$

$$r_{uv} = ?$$

$$x = \frac{5-u}{3} \quad y = \frac{v+7}{2}$$

$$u = 5 - 3x$$

$$r_{xy} \cdot \frac{a}{|a||c|} = r_{uv}$$

$$\text{Here } a = -3 \quad b = 2$$

$$r_{uv} = -r_{xy} = -0.12$$

*Q If x and y are 2 variables, find k such that $x + ky$ and $x + \left(\frac{\sigma_x}{\sigma_y}\right)y$ are uncorrelated

$$u = x + ky \quad v = x + \frac{\sigma_x}{\sigma_y} y$$

$$\text{cov}(u, v) = a \text{var}(x) + (ad + bc) \text{cov}(x, y) + bd \text{var}(y) \quad - (1)$$

also, for uncorrelated $r_{uv} = 0$

$$\Rightarrow \text{cov}(u, v) = 0$$

$$u = ax + by \quad v = cx + dy$$

$$a = c = 1 \quad b = k \quad d = \frac{\sigma_x}{\sigma_y}$$

$$a \text{var}(x) = \text{var}(x) \cdot 1 = \sigma_x^2$$

$$bd \text{var}(y) = k \frac{\sigma_x}{\sigma_y} \sigma_y^2 = k \sigma_x \sigma_y$$

$$ad + bc = \frac{\sigma_x}{\sigma_y} + k$$

$$0 = \sigma_x^2 + \left(\frac{\sigma_x}{\sigma_y} + k\right) r_{xy} \sigma_x \sigma_y + k \sigma_x \sigma_y$$

$$0 = \sigma_x^2 + (\sigma_x + k \sigma_y) r_{xy} \sigma_x + k \sigma_x \sigma_y$$

$$\Rightarrow \sigma_x + (\sigma_x + k \sigma_y) r_{xy} + k \sigma_y = 0$$

$$\Rightarrow \sigma_x (1 + r_{xy}) + k \sigma_y (1 + r_{xy}) = 0$$

$$\Rightarrow (\sigma_x + k \sigma_y) (1 + r_{xy}) = 0$$

$$\Rightarrow k = -\frac{\sigma_x}{\sigma_y}$$

Q $\text{var}(x) = 9$ $\text{var}(y) = 4$ $\text{var}(x-y) = \text{var}(x)$
 $r_{xy} = ?$

$$\text{var}(x-y) = \text{var}(x) - 2\sigma_x\sigma_y r_{xy} + \text{var}(y)$$

$$\text{var}(x) = \text{var}(x) - 2(3)(2)r_{xy} + 4$$

$$0 = 12r_{xy} = 4$$

$$\Rightarrow r_{xy} = 0.33$$

→ Rank co-relation

Sometimes for bivariate data, x & y can't be measured quantitatively but qualitatively done by allotting ranks to individuals. The correlation coeff b/t two series of ranks is called rank correlation coefficient.

If 2 series of ranks are given in characterst x & y then Rank correlation coefficient

$$-1 \leq 1 - \frac{6 \sum d^2}{n^3 - n} \leq 1 \quad n \rightarrow \text{no of observation}$$

$d_i \rightarrow$ diff b/t ranks x_i & y_i

JUDGE-1
Rank

JUDGE-2

Q	x	y	$d = x - y $	d^2
	1	12	11	121
	2	9	7	49
	3	6	3	9
	4	10	6	36
	5	3	2	4
	6	5	1	1
	7	4	3	9
	8	7	1	1
	9	8	1	1
	10	2	8	64
	11	11	0	0
	12	1	11	121



$$n = 12$$

$$n^3 - n = 12(144 - 1) = 12 \times 143 = 1716$$

$$\Rightarrow \text{coeff} = 1 - \frac{6 \times 416}{1716} = 286$$

$$r = 1 - 1.45$$

$$r = -0.45$$

★ R_{xy} close to +1 $\rightarrow$ strictly +ve correlated
 R_{xy} close to -1 $\rightarrow$ strictly -ve correlated

Since R.C.C is negative, we may conclude if rank by judge-1 is high then there is tendency that rank by Judge 2 will be low.

Q In class of 10 students, obtd following marks in Maths & Physics out of 100. Find R.C.C
 Find ranks acc to values

Math	Phy	R_M	R_P	$ R_M - R_P $	$ R_M - R_P ^2$
8	84	10	3	7	49
36	51	7	8	1	1
98	91	1	1	0	0
25	60	9	6	3	9
75	68	4	4	0	0
82	62	3	5	2	4
92	86	2	2	0	0
62	58	6	7	1	1
65	35	5	10	5	25
35	49	8	9	1	1
					<u>90</u>

$$n^3 - n = 1000 - 100 = 900$$

$$\Rightarrow \text{RCC} = 1 - \frac{6 \times 90}{900} = 0.4$$



Q Rank by 3 judges, determine which pair has nearest approach to common taste.

	A	B	C	X	X ²	Y	Y ²	Z	Z ²
$x = A-B $	I	II	III						
$y = B-C $	1	3	6	2	4	3	9	5	25
$z = A-C $	6	5	4	1	1	1	1	2	4
	5	8	9	3	9	1	1	4	16
	10	4	8	6	36	14	16	2	4
	3	7	1	4	16	6	36	2	4
	2	10	2	8	64	8	64	0	0
	4	2	3	2	4	1	1	1	1
	9	1	10	8	64	9	81	1	1
	7	6	5	1	1	1	1	2	4
	8	9	7	1	1	2	4	1	1
				<u>200</u>		<u>205</u>		<u>60</u>	

$$n^3 - n = 900$$

$$(RCC)_{AB} = 1 - \frac{26 \times 200}{3900} = 1 - \frac{5200}{3900} = -0.33$$

strictly negatively correlated

$$(RCC)_{BC} = 1 - \frac{6 \times 205}{900} = -0.36$$

strictly negatively correlated

$$(RCC)_{CA} = 1 - \frac{6 \times 80}{15900} = 1 - \frac{2}{5} = \frac{3}{5} = 0.6$$

strictly positively correlated

⇒ Judges I & III have nearest approach to common taste



→ When 2 observations have same rank

The Rank correlation coefficient

$t \rightarrow$ no of times repeated

$$= 1 - \left[\frac{6 (\sum d^2 + F)}{n^3 - n} \right]$$

$$F = \sum \frac{t^3 - t}{12}$$

Phy	Math	RP	RM	d	d ²
48	13 ⑤	3	5.5	2.5	6.25
33	13 ⑥	5	5.5	0.5	0.25
40	24	4	1	3	9
9	6 ⑧	10	8.5	1.5	2.25
16 ⑦	15	8	4	4	16
16 ⑧	4	8	10	2	4
65	20	1	2	1	1
24	9	6	7	1	1
16 ⑨	6 ⑨	8	8.5	0.5	0.25
57	19	2	3	1	1

$$\sum d^2 = 44, n^3 - n = 900$$

Tab repeat ho, teeno ki sugi leke wohi rank teeno ko deda aur usi walo ko phle ke acc hi deda

$$F = F_{18} + F_{13} + F_6 = \left[\frac{3^3 - 3}{12} \right] + \left[\frac{2^3 - 2}{12} \right] + \left[\frac{2^3 - 2}{12} \right]$$

$$= 2 + 1 = 3$$

$$F = 3$$

$$RCC = 1 - \left[\frac{6 \times 44}{900} \right] = 0.733$$